

10	(a) sharpness: clarity of edges/resolution (of image) contrast: difference in degree of blackening (of structures)	B1 B1	[2]
(b)	(i) X-rays produced when (high speed) electrons hit target/anode <i>either</i> electrons have been accelerated through 80 kV <i>or</i> electrons have (kinetic) energy of 80 keV	B1 B1	[2]
	(ii) $I_T/I = e^{-3.0 \times 1.4}$ $= 0.015$	C1 A1	[2]
(c)	for good contrast, μx or $e^{\mu x}$ or $e^{-\mu x}$ must be very different μx or $e^{\mu x}$ or $e^{-\mu x}$ for bone and muscle will be different than that for muscle so good contrast	B1 M1 A1	[3]